

Draw one number at random from the following set, and let  $X$  be the value of the number that you draw.

$$\{-2, -1, 0, 2\}$$

1. Write down the pmf of  $X$ .

2. Draw the graph of the cdf of  $X$ .

Suppose  $X \sim \text{Poisson}(6)$ , where  $X$  represents the number of students who earn an A+ in Stat 20, and  $Y \sim \text{Poisson}(5)$  represents the number of students who earn an F in stat 20. Since our class is not curved, we can assume that these two quantities are independent.

3. What is the probability that *no one* fails Stat 20? Write an expression for this probability.

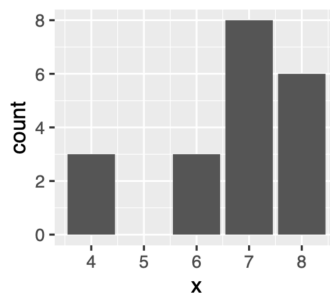
4. What is the probability that *at least* 10 students earn an A+ in Stat 20? Write an expression for this probability.

An American roulette wheel has 18 red, 18 black, and 2 green slots. On each spin, a white ball lands in one of the slots. Suppose you spin the wheel 50 times, and bet on red each time. Let  $X$  be the number of times you win in 50 plays.

5. What is name of the probability distribution of  $X$ , with its parameters?
6. Write an expression for the probability that you win at least 12 times in the 50 spins.
7. Let  $Y$  be a random variable corresponding to the result (win or lose) of the twenty-fifth game. What is the probability distribution of  $Y$ , with its parameter(s)?

Consider a deck of 52 cards. Suppose we are interested in the random variable  $Z$ , which corresponds to the number of hearts dealt in a hand of seven.

8. What is name of the probability distribution of  $Z$ , with its parameters?
9. The following is a visualization which displays twenty draws from a mysterious random variable  $X$ .



- Name a plausible probability distribution based on the draws. Choose from the following three options: *Bernoulli*, *Binomial*, *Poisson*.
- Give a plausible value(s) of the parameter(s) of the probability distribution you chose above.